A Hybrid Movie Recommendation System Based on User-Based and Item-Based Collaborative Filtering

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***Abstract*—Recommendation systems help users discover items that match their interests. Collaborative filtering is a widely used approach that generates recommendations based on user preferences and rating patterns. This project proposes a hybrid movie recommendation system that combines user-based and item-based collaborative filtering techniques. The MovieLens dataset will be used to develop and evaluate the proposed system. A re-ranking approach will be used to reduce popularity bias and give less popular movies a better chance of being recommended. The performance of the individual methods will be compared with the hybrid approach using evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).**

I. Introduction

Today, online platforms offer a large number of movies, making it difficult for users to find movies that match their interests. Recommendation systems help users discover movies based on their preferences and previous ratings. Various recommendation methods have been developed in the literature, including content-based filtering, collaborative filtering, and hybrid approaches [1]. Each method has its own advantages and limitations. Hybrid recommendation systems combine different methods to benefit from their strengths and may improve recommendation performance.

Collaborative filtering is a commonly used approach in recommendation systems. It uses user ratings to identify similar users or similar movies and generate recommendations [2]. User-based collaborative filtering recommends movies based on the preferences of similar users. Item-based collaborative filtering, on the other hand, uses similarities between movies to generate recommendations. However, recommendation systems may tend to recommend popular movies more frequently, while less popular movies may receive less attention. This situation is known as popularity bias.

The aim of this project is to develop a hybrid movie recommendation system that combines user-based and item-based collaborative filtering. The project will also investigate the effects of popularity bias and different user similarity thresholds on recommendation results. The performance of the proposed hybrid model will be compared with the individual approaches.

II. Proposed Method

In this project, a hybrid movie recommendation system will be developed using user-based and item-based collaborative filtering. The main idea is to combine the strengths of both approaches to generate movie recommendations. First, the dataset will be prepared, and similarities between users and movies will be calculated based on their ratings.

In the user-based approach, the system will identify users with similar movie preferences and use their ratings to predict how a user may rate a movie. In the item-based approach, the system will identify movies with similar rating patterns and use the user's previous ratings to make predictions. By combining these approaches, the proposed system aims to use both similarities between users and similarities between movies when generating recommendations. In addition to combining the two collaborative filtering approaches, this project will investigate popularity bias.

Many recommendation algorithms suffer from popularity bias in their output: popular items are recommended frequently and less popular ones rarely, if at all [3]. In other words, recommendation systems may favor movies that have received many ratings, while less popular movies may receive less attention. The aim is not to remove popular movies from recommendations, but to reduce their dominance and give less popular movies a better chance of being recommended. To address this issue, a popularity score will be calculated for each movie based on the number of ratings it has received.

After calculating the popularity scores, a re-ranking approach will be used to reduce the influence of highly popular movies in the recommendation list. The adjusted recommendation score will be calculated using Formula (1):

$$S(u,i) = \hat{r}_{ui} - \lambda P(i) \tag{1}$$

where $\hat{r}_{ui}$ represents the predicted rating for user u and movie i, $P(i)$ represents the movie's popularity score, and λ controls the strength of the popularity penalty. A higher value of λ gives more penalty to popular movies, while a lower value has less effect on their ranking. The impact of this adjustment can be investigated by comparing recommendation accuracy and diversity.

As a further step, the similarity threshold used in the user-based approach may have an important effect on recommendation results. A higher threshold means that only users with very similar movie preferences will be considered. This may lead to more personalized recommendations, but it can also

reduce the number of users available for making predictions. As a result, the system may have difficulty generating recommendations for some users. On the other hand, a lower threshold allows more users with different preferences to participate in the recommendation process. This may provide more rating information, but it can also introduce less relevant preferences and affect prediction accuracy. Therefore, different similarity threshold values may be tested to examine their effects on recommendation accuracy and diversity. The aim is to find a suitable balance between using similar users and including enough rating information to generate reliable recommendations.

III. Dataset And Evaluation

The MovieLens dataset will be used in this project. The dataset contains user ratings, movie information, and demographic details about users. Movie information includes titles and genres, while user information includes age, gender, and occupation. These data provide useful information for analyzing user preferences and generating movie recommendations [4]. These ratings will be used to create a user-movie rating matrix. This matrix will help calculate similarities between users and movies and generate movie recommendations using collaborative filtering methods.

The dataset will be divided into training and testing sets. The training set will be used to calculate similarities between users and movies and build the recommendation models. The testing set will be used to evaluate the models by comparing predicted ratings with actual user ratings. This separation will help measure how well the models perform on data that was not used during training.

The performance of the models will be evaluated using suitable metrics. Metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) may be considered to measure prediction accuracy [5]. Depending on the project's progress, additional metrics related to recommendation quality and diversity may also be investigated. The results of user-based, item-based, and hybrid models will be compared to examine their performance.

IV. Project Timeline

- **Nov. 05 – Nov. 12:** Dataset preparation. *Expected outcome:* The MovieLens dataset will be cleaned and organized. A user-movie rating matrix will be created to represent the ratings given by users to movies.
- **Nov. 13 – Nov. 23:** Development of user-based and item-based collaborative filtering models. *Expected outcome:* User-based and item-based recommendation models will be developed. The similarity matrices will be calculated to generate movie recommendations based on user preferences and movie similarities.
- **Nov. 24 – Dec. 05:** Development of the hybrid model. *Expected outcome:* User-based and item-based predictions will be combined to create a hybrid model that generates movie recommendations.
- **Dec. 06 – Dec. 15:** Implementation of popularity bias adjustment and similarity threshold experiments. *Expected outcome:* A popularity score will be calculated for each movie, and a re-ranking approach will be applied. Different similarity threshold values will also be tested to examine their effects on recommendations.
- **Dec. 16 – Dec. 22:** Model testing and performance evaluation. *Expected outcome:* Hybrid models will be evaluated using MAE, MSE, and RMSE. Prediction accuracy and recommendation diversity will be compared.

V. Backup Plan

If the hybrid model cannot be implemented as planned, user-based and item-based collaborative filtering models may be evaluated separately. Alternatively, with the instructor's approval, the latent factor model may be considered as an alternative recommendation approach. The most appropriate option will be determined based on the project's progress and the time remaining. If the popularity bias correction or similarity threshold experiments cannot be completed, the project will focus on the basic recommendation models and their performance evaluation. In this case, the completed model will still be compared using metrics such as MAE, MSE, RMSE, and similar measures.

VI. Expected Results

The expected outcome of this project is a hybrid movie recommendation system that combines user-based and item-based collaborative filtering. The system is expected to provide movie recommendations based on users' preferences and rating patterns. The performance of the hybrid model will be compared with the individual user-based and item-based approaches. The aim is to investigate whether combining these methods can improve recommendation accuracy.

In addition, the project aims to examine the effect of popularity bias on recommendation results. The re-ranking approach is expected to help reduce the dominance of popular movies and provide more diverse recommendations. The effects of different similarity threshold values will also be investigated to understand how they influence recommendation accuracy and diversity.

References

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